

QUANT HACKATHON



Dates: 14th to 16th March

Momentum Trading Strategy using Machine Learning

QuantQuest Hackathon 3 IIT Mandi (E-Summit 2026)

Paras Surve

Objective



Project Goal

Develop a systematic momentum trading strategy enhanced by machine learning to identify and capitalise on price trends in large-cap US equities.

Approach

Combine traditional momentum indicators with Random Forest classification to predict weekly return probabilities and construct an optimised portfolio.



Dataset Overview

Data Source

YahooFinancedaily OHLCV
data

201732025 period

Stock Universe

10large-cap USstocks

Technology, finance, healthcare
sectors

Stock Selection

- AAPL, MSFT, GOOGL, AMZN, META
- TSLA, JPM, V, JNJ, BRK.B

Feature Engineering

Five technical indicators extracted from daily price data to capture momentum and volatility dynamics:

01

1-Day Return

Logarithmic price change from previous trading day

02

5-Day Momentum

Cumulative return over past 5 trading days

03

10-Day Momentum

Intermediate-term trend indicator

04

20-Day Momentum

Longer-term directional bias signal

05

Rolling Volatility

20-day standard deviation of returns

Machine Learning Model

Random Forest Classifier

Ensemble tree-based model selected for robustness to non-linear relationships and feature importance analysis.

Target Variable

Binary classification predicting probability of positive next-week return based on current feature values.

Advantages

- Handles non-linear patterns effectively
- Provides probability estimates
- Resistant to overfitting
- Interpretable feature importance



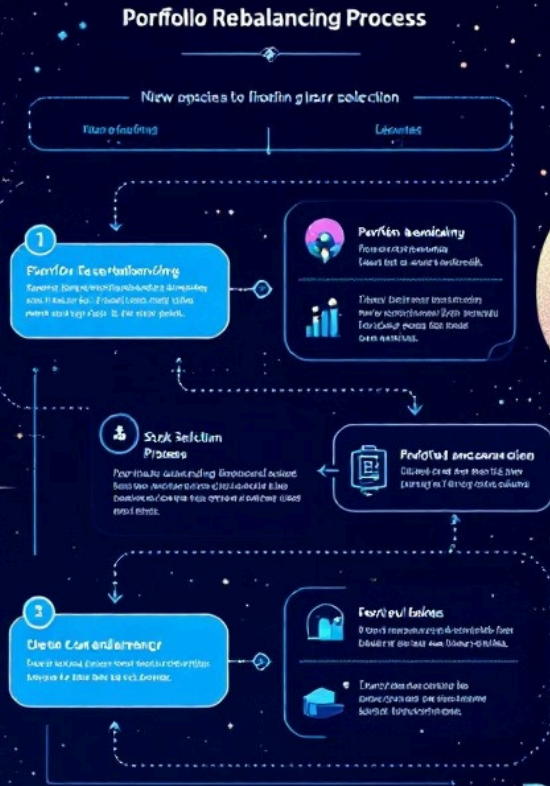
- 1
- 2
- 3
- 4

Calculate momentum scores for all 10 stocks

Identify top 2 stocks with highest momentum signals

Allocate 50% to each selected stock

Adjust positions weekly based on updated rankings



Backtesting Setup

1

Training Period

2017-2022 (5 years)

Model fitting and hyperparameter tuning

2

Testing Period

2023-2025 (3 years)

Out-of-sample performance evaluation

3

Transaction Costs

0.1% entry fee

0.1% exit fee

Realistic slippage assumptions

Performance Results

4.0x

3-year total performance vs.

benchmark

20%

Compound annual growth rate
achieved

0.89

Risk-adjusted performance

metric

-31%

Largest peak-to-trough
decline

Strategy demonstrated consistent outperformance with acceptable downside risk, generating 4.0x cumulative returns over 3 years with Sharpe ratio of 0.85.

Transaction cost included:



Cumulative Returns Performance

Strategy outperformed buy-and-hold benchmarks significantly over testing period

- ❏ The equity curve shows consistent upward trajectory with periodic drawdowns, demonstrating the strategy's ability to capture momentum while managing downside exposure.

Conclusion

Key Findings

Machine learning-based momentum strategy successfully captured price trends in large-cap US stocks from 2023 to 2025, achieving 4.0x cumulative returns

with strong risk-adjusted performance.

Success Factors

- Effective feature engineering combining multiple momentum horizons
- Random Forest classifier capturing non-linear relationships
- Systematic rebalancing protocol



Future Enhancements

Consider incorporating sentiment analysis, macroeconomic indicators, and deep learning architectures for improved predictive power.